

Gender Pricing and Service Duration: Evidence from European Salons

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Abstract

How is the gender price gap structured within posted menus? Using within-salon fixed effects on 33,682 haircut listings from European salons, I decompose the within-salon gap into flat-premium and slope components. The differential mainly takes the form of a steeper posted price–duration schedule (€0.061 per listed minute higher for female-coded services; €0.100 in unisex salons) rather than a flat surcharge. In unisex salons, the remaining flat premium is €0.741, while the slope pattern persists in single-employee, wet-cut, and matched-title restrictions, holds in all ten countries separately, and is also present for children’s haircuts. Because salons choose listed duration jointly with price and label, I interpret these estimates as posted-schedule differences rather than a causal return to time. That structure matters because it is less salient to consumers and harder for surcharge-focused regulation to detect.

Keywords: gender pricing, haircut markets, service markets, posted prices

JEL Classifications: D18, J16, L84, D40

1 Introduction

In product markets, detailed analyses find that cost differences rather than markups explain observed gender price gaps, consistent with the ease of arbitrage in posted-price retail (Moshary et al., 2023; Barnes and Brounstein, 2024). In service markets, where non-transferability precludes arbitrage (Varian, 1989; Stole, 2007), gender-based pricing has long been documented (Liston-Heyes and Neokleous, 2000; Duesterhaus et al., 2011; Heiden and Wersig, 2018; New York City Department of Consumer Affairs, 2015), but existing work does not characterize *how* the differential is structured. This distinction matters for three reasons. First, a flat premium imposes the same burden regardless of length, whereas a steeper schedule makes the gap grow with service duration. Second, the common defense that women’s services simply take longer cannot, on its own, account for a steeper posted price–duration schedule, since longer services would shift total price but not the slope. Third, enforcement aimed at visible surcharges can miss a differential embedded in the schedule.

This paper uses 33,682 haircut listings from 13,658 salons in ten European countries to decompose the within-salon gender gap into a flat premium and a slope. Female-coded services follow a steeper posted price–duration schedule: the interaction term is €0.061 per listed minute in the full sample and €0.100 in unisex salons. In unisex salons, once the interaction is included, the remaining flat premium falls from €4.534 to €0.741. The same pattern appears in single-employee salons, wet-cut subsamples, matched-title restrictions, all ten countries, and children’s cuts. This descriptive characterization is itself new: prior work establishes that women pay more in salons, but does not show whether the within-salon gap is posted as an intercept shift or a steeper schedule. Because listed duration is itself a salon choice rather than an exogenous measure of service time, I read these estimates as a description of how posted schedules differ by gender rather than as a causal estimate of per-minute discrimination.

2 Empirical Framework and Data

2.1 Data

The main regression sample contains 33,682 adult haircut listings from 13,658 salons in Austria, Belgium, France, Germany, Italy, the Netherlands, Portugal, Spain, Switzerland, and the United Kingdom, collected in June 2025 from a major European online booking platform. The sample is restricted to haircut listings, excluding bundled treatments such as coloring, perms, extensions, and beard work. Each listing records a service name, duration, price, and a platform gender tag. Prices are converted to euros using OECD PPP data (Organisation for Economic Co-operation and Development, 2025). Female-coded haircuts average €34.08 versus €25.83 for male-coded haircuts.

2.2 Empirical Strategy

I approximate each salon’s posted menu with a linear price–duration schedule that allows both the intercept and slope to differ by gender. The estimating equation with salon fixed effects is:

$$\text{price}_{ij} = \beta_1 \cdot \text{female}_{ij} + \beta_2 \cdot \text{duration}_{ij} + \beta_3 \cdot (\text{female}_{ij} \times \text{duration}_{ij}) + \alpha_j + \varepsilon_{ij}$$

where i indexes services within salon j , and α_j are salon fixed effects. Standard errors are clustered at the salon level. With salon fixed effects, β_1 and β_3 are identified from within-salon comparisons of male- and female-coded listings. Because identification is within salon, salon fixed effects absorb time-invariant country and regional pricing differences. The pooled coefficients therefore average within-salon schedule differences across countries; Appendix B reports the preferred unisex-salon interaction coefficient separately for each country, and all ten are positive.

Here, “duration” is the posted duration chosen by the salon jointly with the price and gender label. As an external benchmark, Chen et al. (2010) measure average durations of 35.6 minutes for men’s haircuts and 51.4 minutes for women’s haircuts in Taiwanese salons, close to the unfiltered raw means of 35.9 and 53.3 minutes. Listed durations therefore appear anchored to

plausible service times on average, even though duration remains a choice variable rather than an exogenous measure of service time. Accordingly, β_3 does not identify the marginal price of time, and the gender interaction does not identify a discriminatory price per minute. Instead, it captures whether salons post systematically steeper price–duration schedules for female-coded than for male-coded services. The subsequent restrictions test how far that pattern persists as differences in provider assignment, service technology, and menu labels are narrowed.

3 Results

3.1 Adult Haircut Pricing

Table 1 reports adult haircut results. In the full sample, female-coded services are €4.648 more expensive conditional on listed duration (Column 1). Once the interaction is added, the slope term is 0.061 (Column 2, $p < 0.01$), indicating that posted prices rise faster with listed duration for female-coded than for male-coded services within the same salon.

Columns (3) and (4) restrict the sample to unisex salons. This removes comparisons between cheaper male-only barbershops and mixed salons. In the interaction specification, the female intercept falls to €0.741 and is only marginally significant, while the slope differential rises to 0.100 ($p < 0.01$). At 60 listed minutes, this implies an approximately €6.00 higher posted price within the same salon menu.

3.2 Robustness Checks

Table 2 reports four restrictions, each holding a different potential confounder fixed. In single-employee salons, which hold provider assignment fixed within salon, the interaction remains 0.103 ($p < 0.01$) and the intercept is small and statistically insignificant. In wet-cut subsamples, which narrow production-technology differences, the interaction is 0.127 ($p < 0.01$). The children’s exercise also yields a positive slope differential of 0.069 ($p < 0.01$). Columns (5)–(6) use a narrower paired-title sample that requires male and female listings within a salon to have identical nongendered service titles with additional descriptive content. This sample contains 1,548 listings from 725 salons and is dominated by wash–cut–blow-dry titles, which

Table 1: Adult Haircuts: Posted Price–Duration Schedule Differences

<i>Dependent variable: price</i>				
	All Salons		Unisex Salons Only	
	(1)	(2)	(3)	(4)
Duration (min)	0.528*** (0.007)	0.485*** (0.009)	0.536*** (0.008)	0.462*** (0.011)
Female	4.648*** (0.149)	2.295*** (0.398)	4.534*** (0.156)	0.741* (0.431)
Duration $\times$ Female		0.061*** (0.010)		0.100*** (0.012)
Salon Fixed Effects	Yes	Yes	Yes	Yes
Observations	33682	33682	20758	20758
Number of salons	13658	13658	6459	6459
R^2	0.603	0.605	0.621	0.625

Notes: The dependent variable is price. Standard errors, clustered at the salon level, are reported in parentheses. Columns (1)-(2) present results for the full sample of salons. Columns (3)-(4) are restricted to unisex salons that offer services to both men and women. Statistical significance is denoted as: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

account for just over half of rows.¹ If hidden bundle differences under coarse labels were driving the baseline slope differential, the estimate should attenuate as labels are forced to match. Instead, the interaction rises to 0.252 ($p < 0.01$) and the female intercept is small and statistically insignificant. Taken together, the matched-title and wet-cut restrictions narrow scope for an explanation based on residual complexity: matched titles hold nominal service description constant, and the wet-cut restriction holds the cutting technology constant. In both cases the slope differential persists or grows.

3.3 Interpretation

The estimates point to three implications.

Heterogeneous posted differential. In the preferred unisex sample, the 0.100 interaction implies about €3.00 at 30 listed minutes and €6.00 at 60 minutes. A slope-based schedule therefore scales the posted gap with appointment length rather than imposing a uniform flat

¹A simple salon-level selection check shows that matched-title salons are not obviously smaller or longer-duration establishments than other unisex salons: mean employee count is 3.0 versus 3.3 and mean average listed duration is 42.1 versus 41.7 minutes. They are, however, more concentrated in Germany and have higher pooled mean male-cut prices (€32.5 versus €23.7), so I interpret the matched-title restriction as a selected robustness check rather than a representative subsample.

Table 2: Robustness Checks

<i>Dependent variable: price</i>						
	Single Employee		Wet Cuts		Matched Titles	
	(1)	(2)	(3)	(4)	(5)	(6)
Duration (min)	0.556*** (0.016)	0.481*** (0.020)	0.437*** (0.018)	0.343*** (0.027)	0.304*** (0.031)	0.122*** (0.040)
Female	4.134*** (0.282)	0.306 (0.814)	6.496*** (0.335)	1.356 (0.993)	9.814*** (0.626)	-0.632 (1.452)
Duration $\times$ Female		0.103*** (0.021)		0.127*** (0.025)		0.252*** (0.035)
Salon Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6127	6127	7456	7456	1548	1548
Number of salons	1881	1881	3680	3680	725	725
R^2	0.605	0.608	0.594	0.599	0.681	0.701

Notes: The dependent variable is price. Standard errors, clustered at the salon level, are reported in parentheses. Columns (1)-(2) restrict the sample to owner-operated or single-employee salons that offer services to both men and women, thereby holding the service provider constant within salon. Columns (3)-(4) restrict the sample to wet-cut services from unisex salons. Columns (5)-(6) restrict the sample to within-salon male/female listings with identical nongendered service titles and additional descriptive content. Statistical significance is denoted as: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

premium.

Robustness across subsample restrictions. No single restriction identifies the slope differential, but each rules out a different alternative explanation: unisex salons remove broad between-salon composition, single-employee salons narrow provider assignment, matched titles narrow nominal service composition, wet cuts narrow cutting technology, and the children's exercise shows the same schedule structure in a distinct subsample. Residual complexity differences within nominally similar services cannot be ruled out fully, yet the combined pattern is difficult to reconcile with complexity or bundling as the dominant explanation.

Consumer and policy salience. Embedded in the slope rather than a flat surcharge, the differential is less visible to consumers and harder to detect in menu comparisons focused on entry prices. A customer can spot a visible entry premium immediately, but a schedule-based gap requires comparing several services and durations. Rules aimed at explicit surcharges can therefore miss the form in which the differential is actually posted. This implication holds regardless of which mechanism generates the slope differential. California's Gender Tax Repeal Act, for example, prohibits gender-based pricing for similar services but explicitly permits

price differences based on the amount of time, difficulty, or cost of providing the service, making schedule-based differentials especially salient in practice.

4 Conclusion

Within salons, gender-labeled haircut menus differ mainly through the slope of the posted price–duration schedule rather than a flat premium. This pattern holds across unisex, single-employee, wet-cut, and matched-title restrictions, and across all ten countries in the sample. The contribution is to show that the within-salon gap is embedded in posted minutes rather than an entry surcharge — a descriptive fact that matters because schedule-based differentials are less visible to consumers and harder for surcharge-focused rules to detect. The estimates cannot separate discrimination from complexity, bundling, or demand differences; what they show is the form in which the gap is posted.

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Disclosure of interest

The author reports there are no competing interests to declare.

Data Availability

The data were collected from a commercial booking platform whose terms of service prohibit redistribution. The web scraping and analysis code is publicly available at <https://github>.

[com/RichSchulz/gender-pricing-service-duration-replication](https://richschulz.com/RichSchulz/gender-pricing-service-duration-replication).

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A Children's Haircuts

Table 3 reports a supplementary children's exercise. The headline specification estimates a €2.16 differential for girls ($p < 0.01$). Once the interaction is added, the girl's-cut intercept falls to €0.063 and the slope term is 0.069 ($p < 0.01$), implying an additional €4.14 at 60 listed minutes. The same schedule structure therefore appears for children's haircuts as well: little flat premium once the interaction is included, but a positive slope differential.

Table 3: Children's Haircuts: Posted Price–Duration Schedule Differences

	<i>Dependent variable: price</i>	
	(1)	(2)
Duration (min)	0.373*** (0.013)	0.321*** (0.025)
Girl's Cut	2.160*** (0.159)	0.063 (0.615)
Unisex Cut	-0.271 (0.302)	-1.712** (0.842)
Duration × Girl		0.069*** (0.021)
Duration × Unisex		0.049* (0.030)
Salon Fixed Effects	Yes	Yes
Observations	6947	6947
Number of salons	4030	4030
R^2	0.412	0.414

Notes: The dependent variable is price. Standard errors, clustered at the salon level, are reported in parentheses. The reference category for gender is Boys' Cuts. Statistical significance is denoted as: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

B Country-Specific Unisex Estimates

Table 4 reports the preferred unisex-salon interaction coefficient separately for each country. The point estimate remains positive in all ten countries, indicating that the pooled slope differential is not driven by a single country's pricing structure. Eight of the ten coefficients are statistically significant at the 10% level, and seven are significant at the 5% level.

Table 4: Country-Specific Unisex-Salon Slope Estimates

Country	Duration $\times$ Female	SE	Observations	Number of salons
Austria	0.118	(0.041)	1285	312
Belgium	0.371	(0.110)	168	40
Switzerland	0.125	(0.052)	925	230
Germany	0.066	(0.025)	4482	1153
Spain	0.143	(0.038)	1551	483
France	0.005	(0.059)	1673	514
United Kingdom	0.075	(0.034)	2986	1054
Italy	0.139	(0.020)	6379	2328
Netherlands	0.092	(0.060)	946	216
Portugal	0.133	(0.070)	363	129

Notes: Each row reports the interaction coefficient from the preferred unisex-salon specification estimated separately by country. Standard errors are clustered at the salon level.